

2021 Ph H2 Q15

Section: Our Dynamic Universe

Topic: Terminal Velocity & Viscosity

Ball bearings fall through glycerol. (a) Explain forces leading to terminal velocity. (b)(i) Plot graph of v_t against d^2 . (b)(ii) Explain why systematic uncertainty may be suspected. (b)(iii) Calculate gradient. (b)(iv) Use gradient to calculate viscosity of glycerol.

Worked solution

(a) Initially the ball accelerates because weight $>$ drag + upthrust. As speed increases, resistive force (viscous drag) increases. Eventually, resistive force + upthrust = weight. Net force = 0, so acceleration = 0. The ball then falls at constant terminal velocity.

(b)(i) A graph of v_t (y-axis) against d^2 (x-axis) should be plotted. The data points lie close to a straight line through the origin.

(b)(ii) If the line does not pass exactly through the origin, this suggests a systematic error. Likely source: error in measurement of diameters, or timing error (reaction time delay) affecting all velocity measurements in the same way.

(b)(iii) Gradient = $\Delta v_t / \Delta(d^2)$.

Calculated best-fit slope = $3.32 \times 10^3 \text{ (m/s)/(m}^2\text{)}$.

Answer: 3.1×10^3

(b)(iv) From Stokes' law: $v_t = (g d^2)/(18\eta)$.

So slope = $v_t/d^2 = g/(18\eta)$.

$\eta = g/(18 \times \text{slope}) = 9.8/(18 \times 3.32 \times 10^3) = 0.00 \text{ Pa}\cdot\text{s}$

Answer: 0.18 Pa·s

Final answers

(a) Terminal velocity reached when resistive + upthrust = weight

(b)(i) Straight line graph through origin

(b)(ii) Systematic error: e.g. measurement bias

(b)(iii) Gradient $\approx 3.1 \times 10^3$

(b)(iv) $\eta \approx 0.18 \text{ Pa}\cdot\text{s}$

Revision tips

- Terminal velocity: balance of forces $\rightarrow$ no acceleration.
- Viscous drag $\propto$ speed and cross-sectional area.
- Plot v_t vs d^2 for straight-line relationship.
- Systematic errors shift all results (e.g. instrument zero error).
- Viscosity from gradient: slope = $g/(18\eta)$.